

CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Schiebinger

Generated on 2026-10-05 18:10

Run		Perturbations (perturbation_simulation)	
Split	train	✓	P1 OV Obox6
Stimulus penalisation	1.0	✓	P2 OV Zfp42
Prior penalisation	1.0		
Genes	108		
Stimuli	2		
Observed cells	5618		
Timepoints	0, 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, 144, 156, 168, 180, 192, 204, 216, 228, 240, 252, 264, 276, 288, 300, 312, 324, 336, 348, 360, 372, 384, 396, 408, 420, 432		
Cell types	Stromal, MEF, Neural, Trophoblast, Epithelial, iPSC		
Network shown	inter_simul.npy, network #0 / 1		
UMAP preprocessing	normalise=False, log1p=True		

Contents

1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots) and top-10 regulators + stimulus
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth
5. Learned dynamics — mRNA and protein velocity fields (mechanistic and along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as $NB(k, c / s_i)$ when the depth factor s_i is applied.

Depth

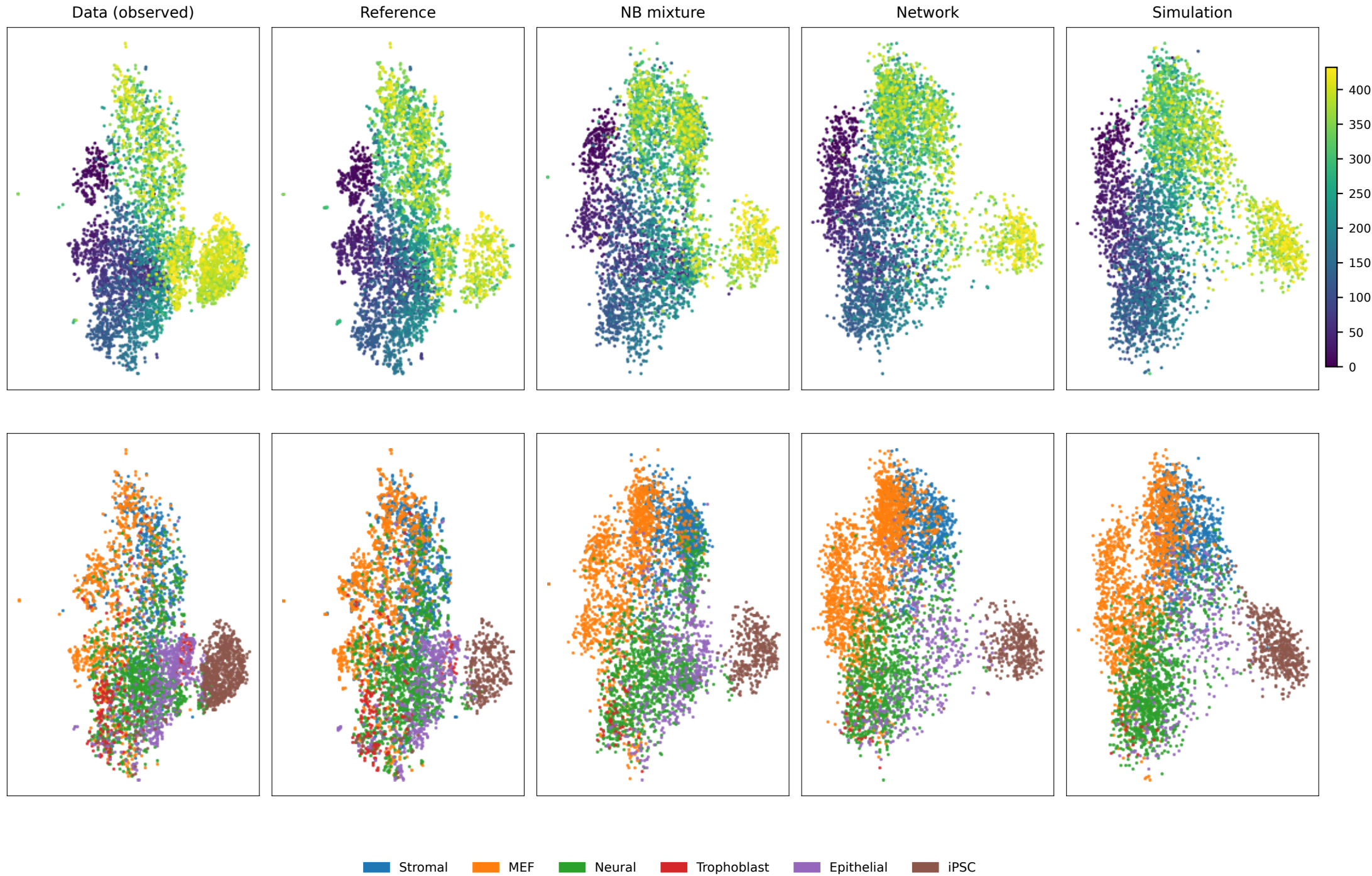
not estimated

Reason

108 genes < 10000

1. Generative model — trajectories and simulation

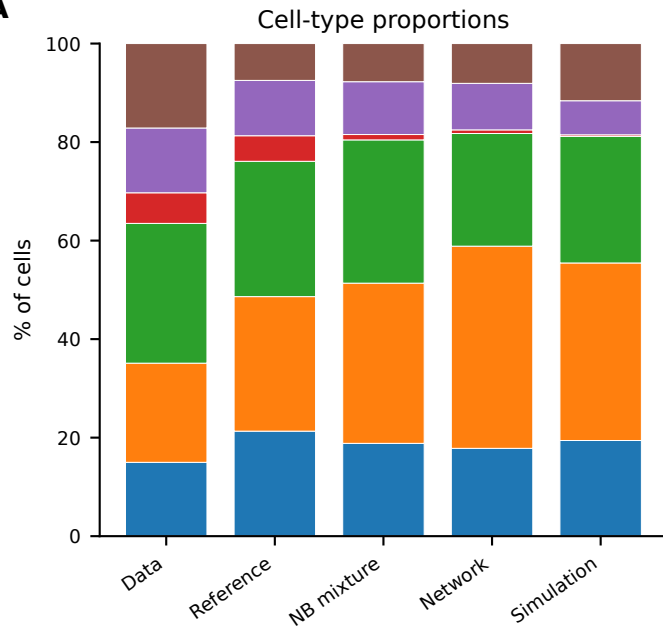
Joint UMAP of the observed data, the trajectories (Reference: one state per ancestor, i.e. without proliferation, as the simulation), NB mixture and network-driven modes along them, and the full simulation.



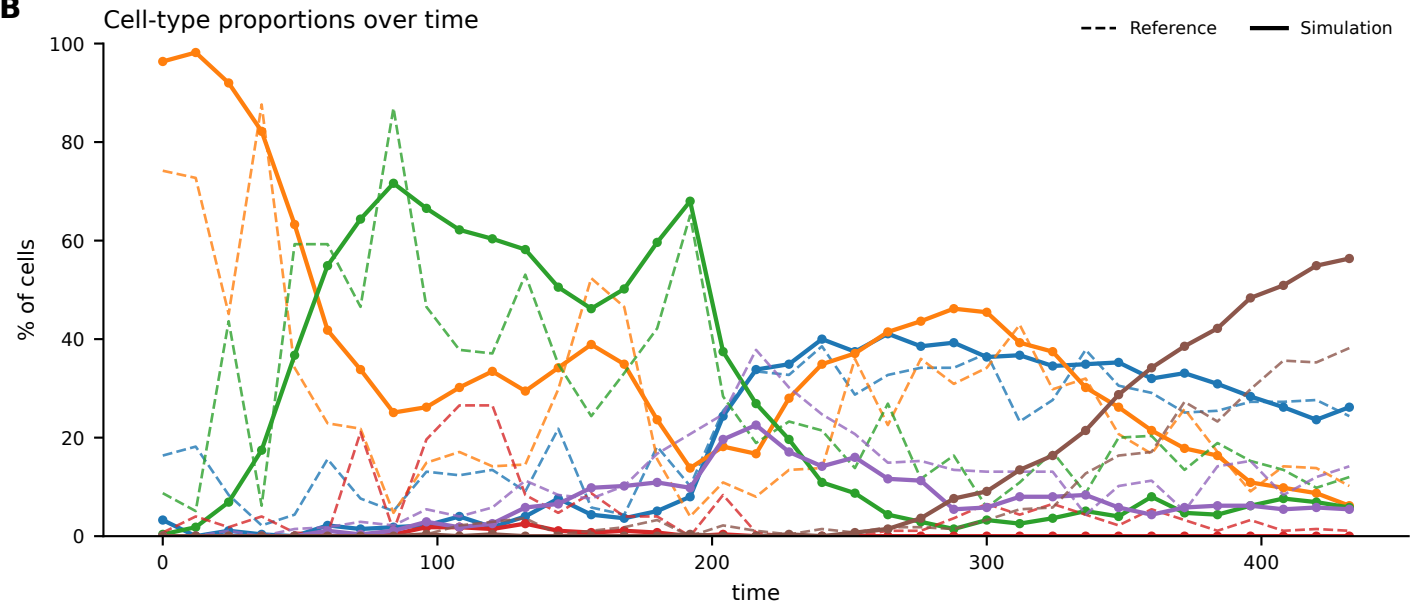
1. Generative model — quantitative checks

Reference = RNA of the trajectories equivalent to the simulation (without proliferation).

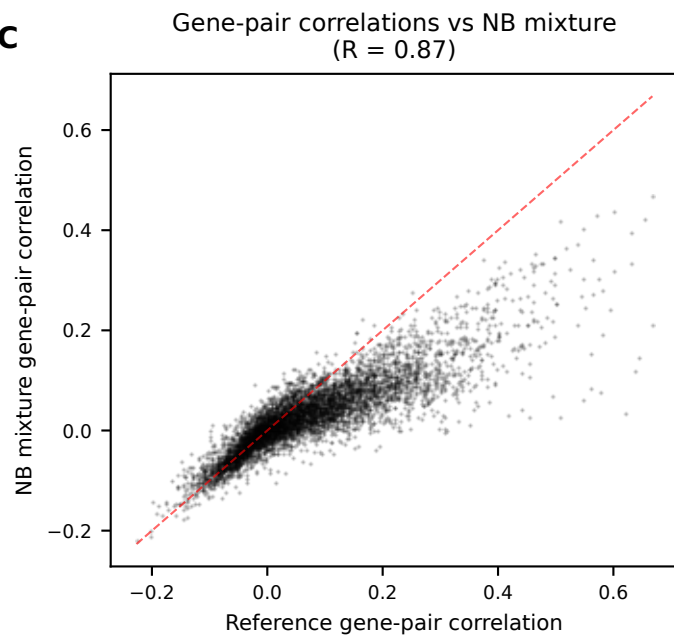
A



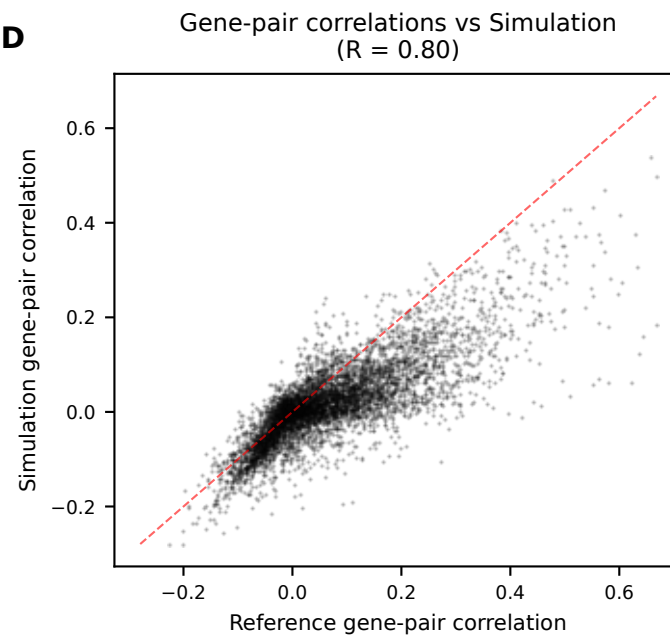
B



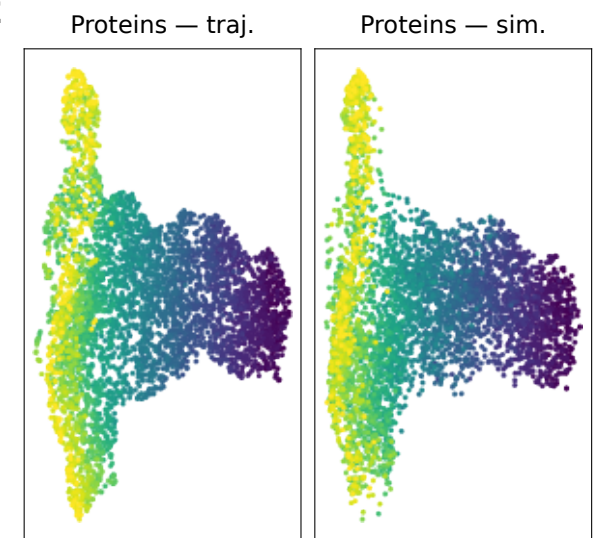
C



D



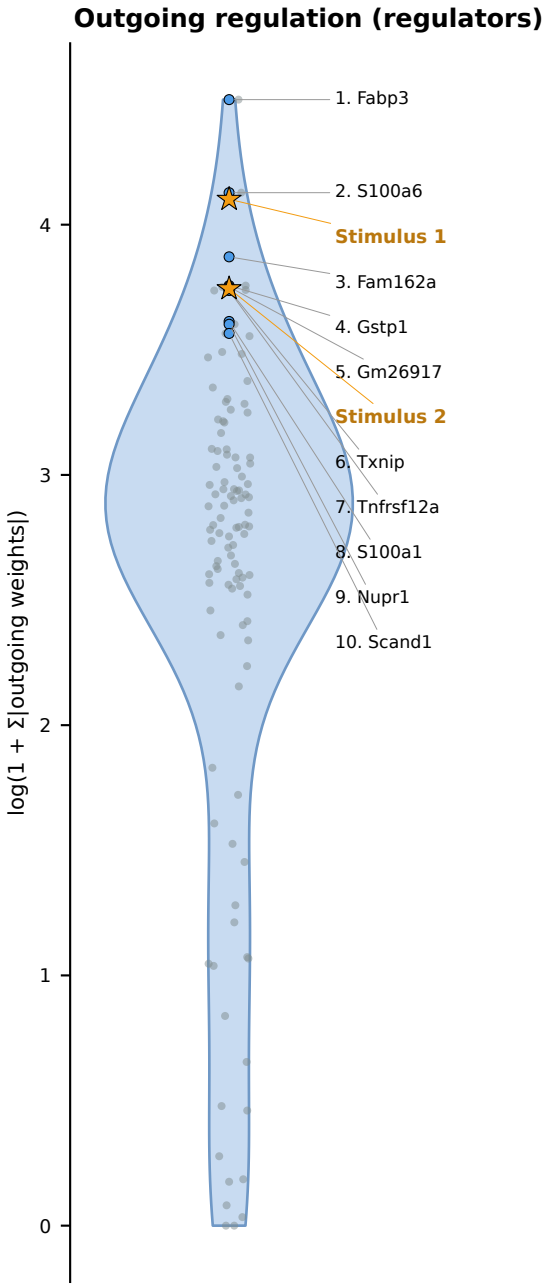
E



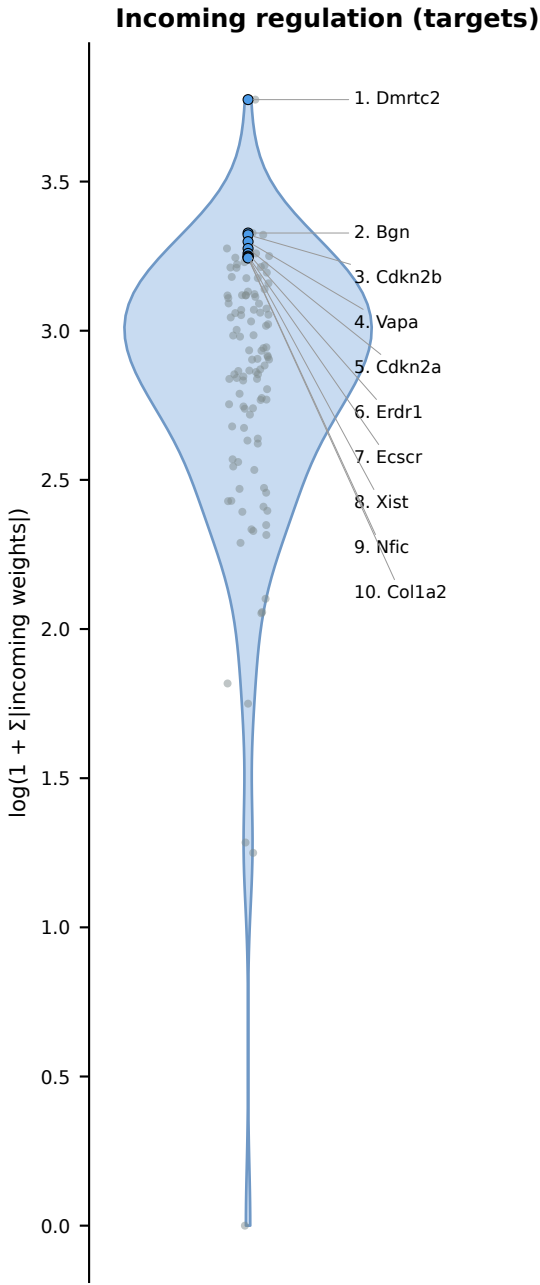
2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 10 genes labelled; stimulus shown as a star (stimulus = 1.0 $\geq$ 0.5).

A



B

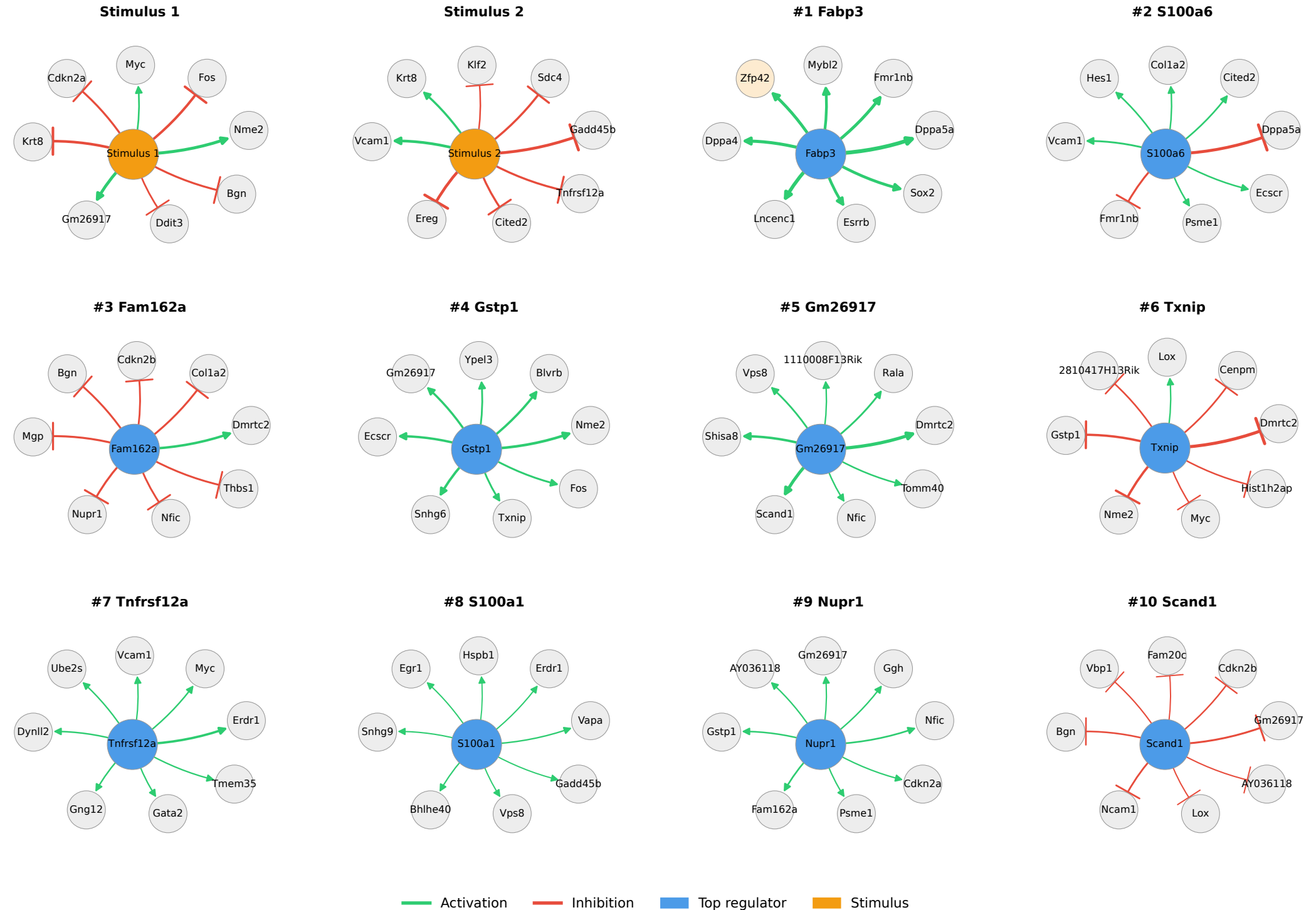


C

Top regulators				
#	Regulator	$\Sigma w $	# act.	# inh.
1	Fabp3	89.00	56	30
2	S100a6	61.05	84	11
3	Fam162a	47.01	32	52
4	Gstp1	41.80	50	11
5	Gm26917	41.58	54	16
6	Txnip	41.06	59	39
7	Tnfrsf12a	40.97	79	11
8	S100a1	36.09	99	3
9	Nupr1	35.67	72	23
10	Scand1	34.35	31	58
★	Stimulus 1	59.44	43	47
★	Stimulus 2	41.29	39	25

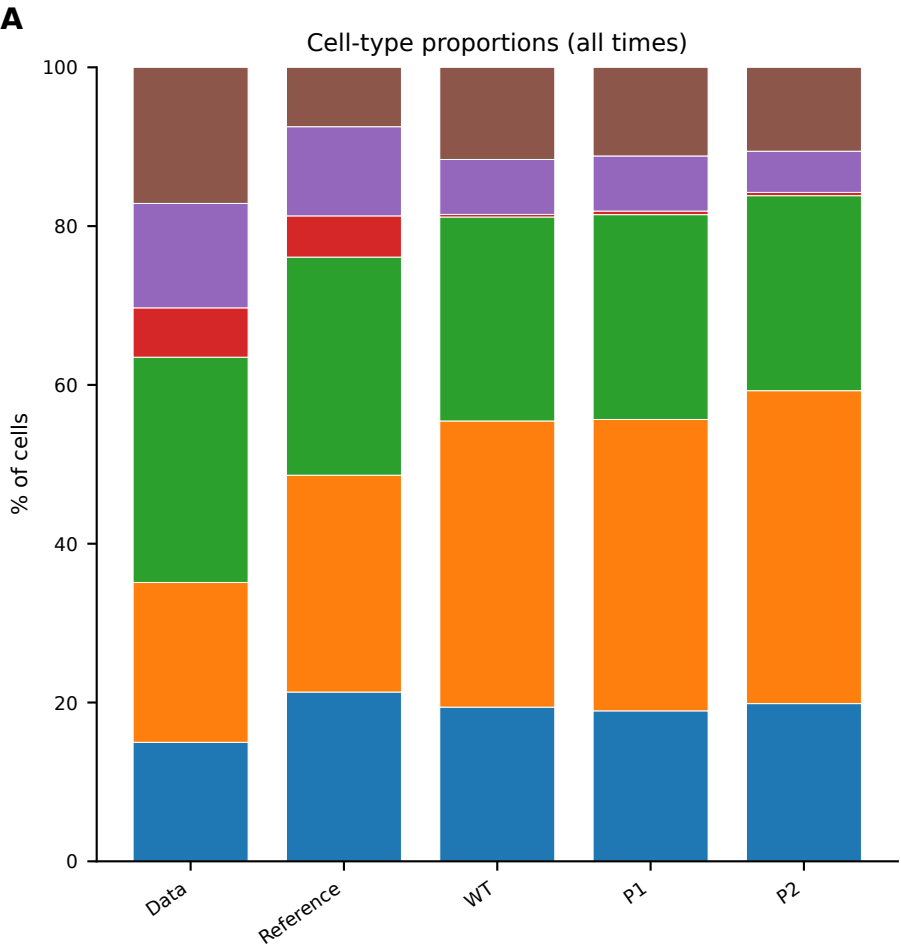
2. Gene regulatory network — top regulators and their main targets

Top 8 targets per regulator; edge width $\propto$ |weight| (global scale). Perturbed genes (KO/OV) are highlighted in beige.

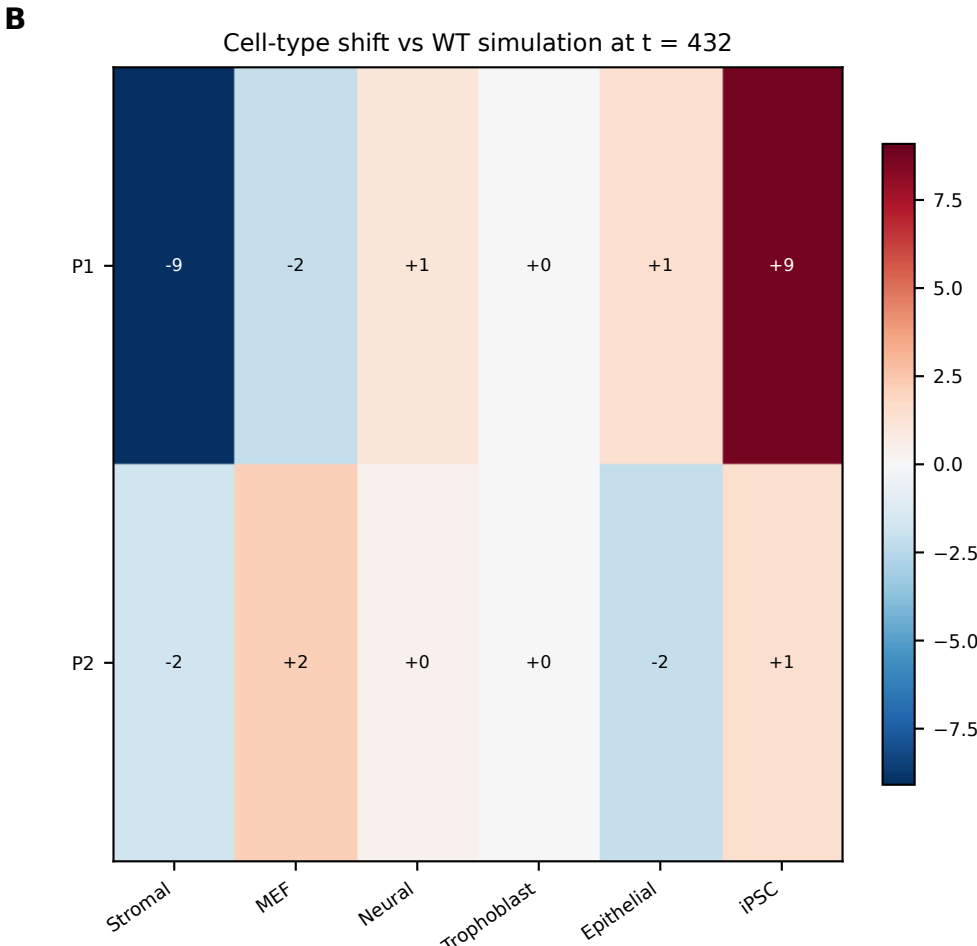


3. In-silico perturbations — overview

Cell types predicted by a random forest trained on the observed data.



P1 OV Obox6

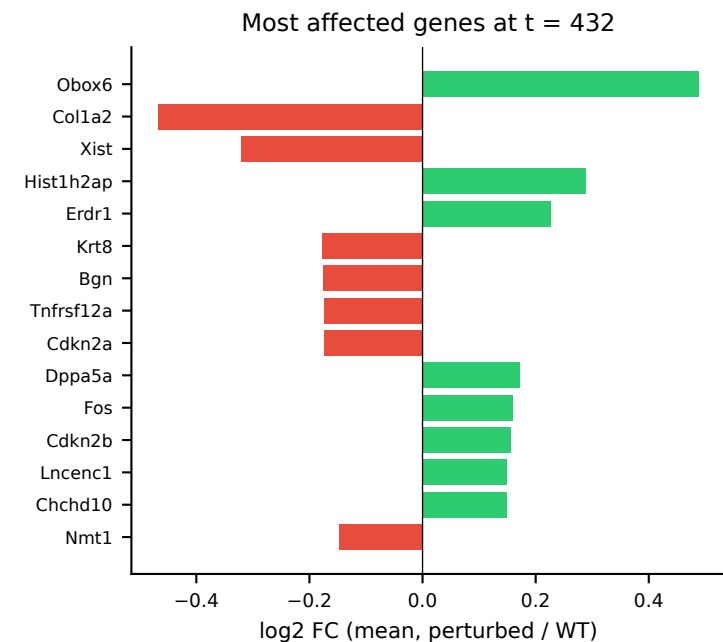
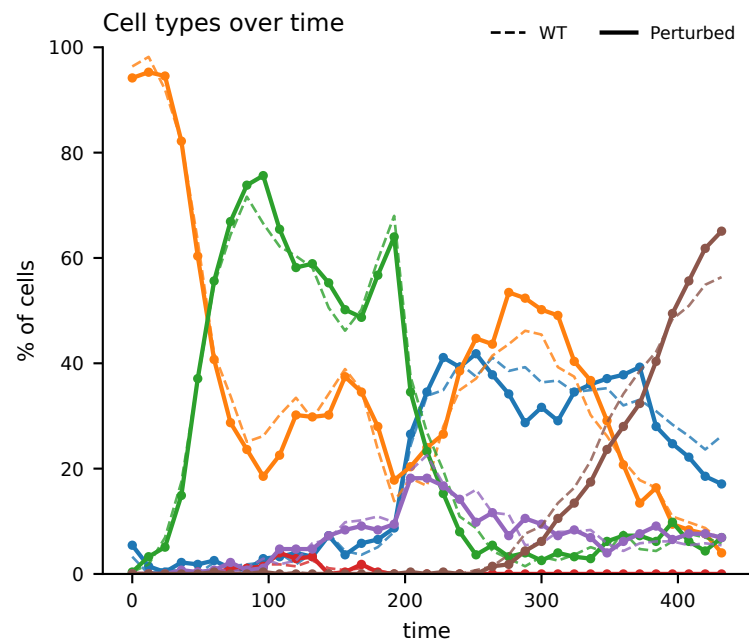
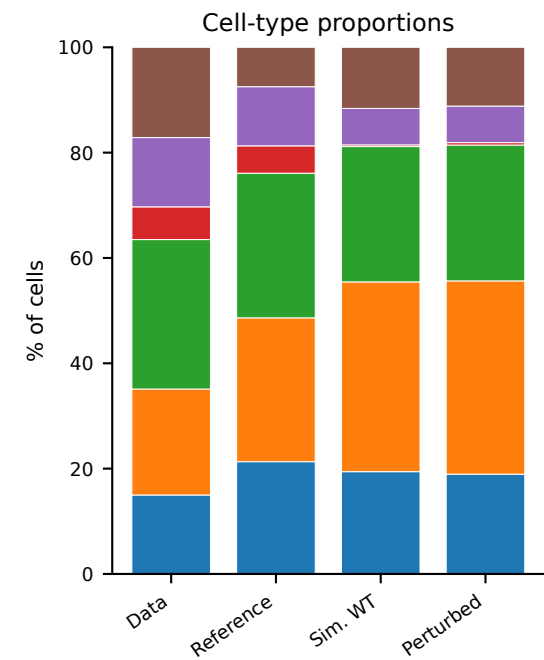
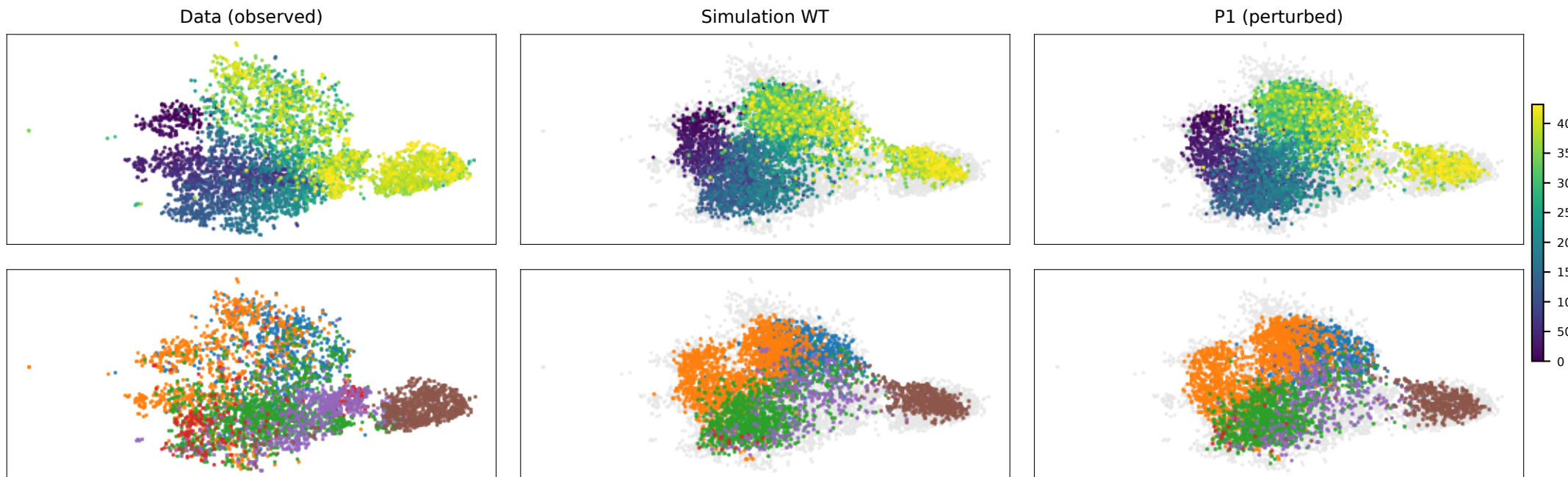


P2 OV Zfp42

Stromal MEF Neural Trophoblast Epithelial iPSC

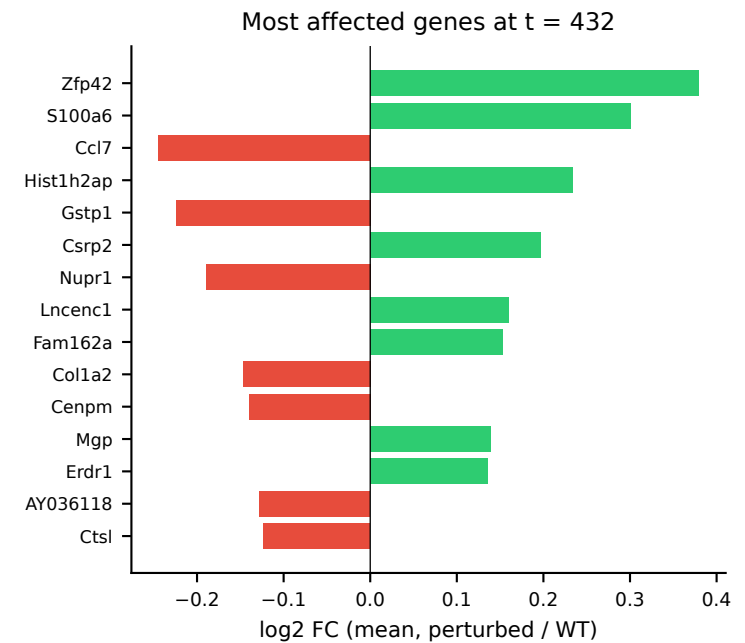
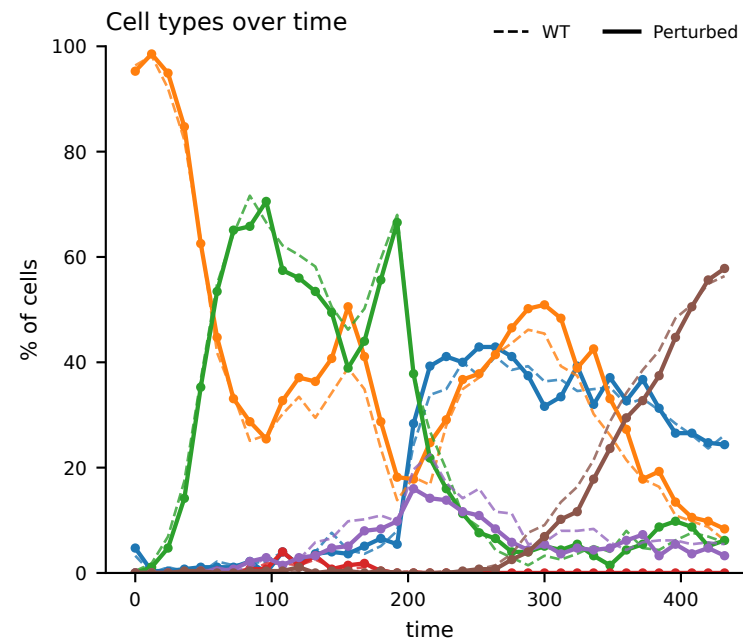
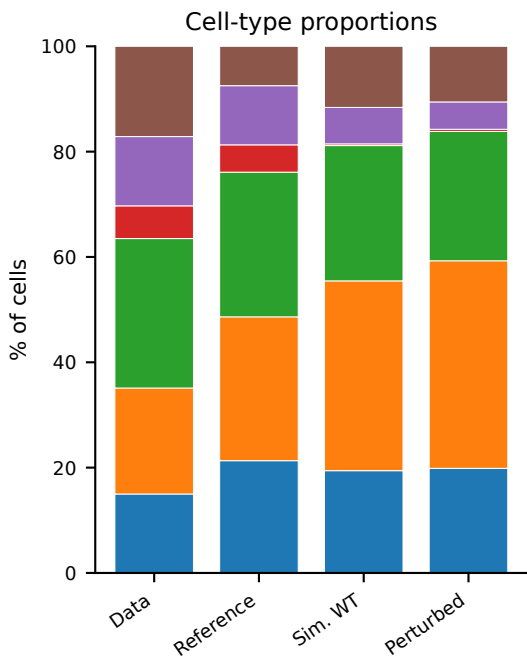
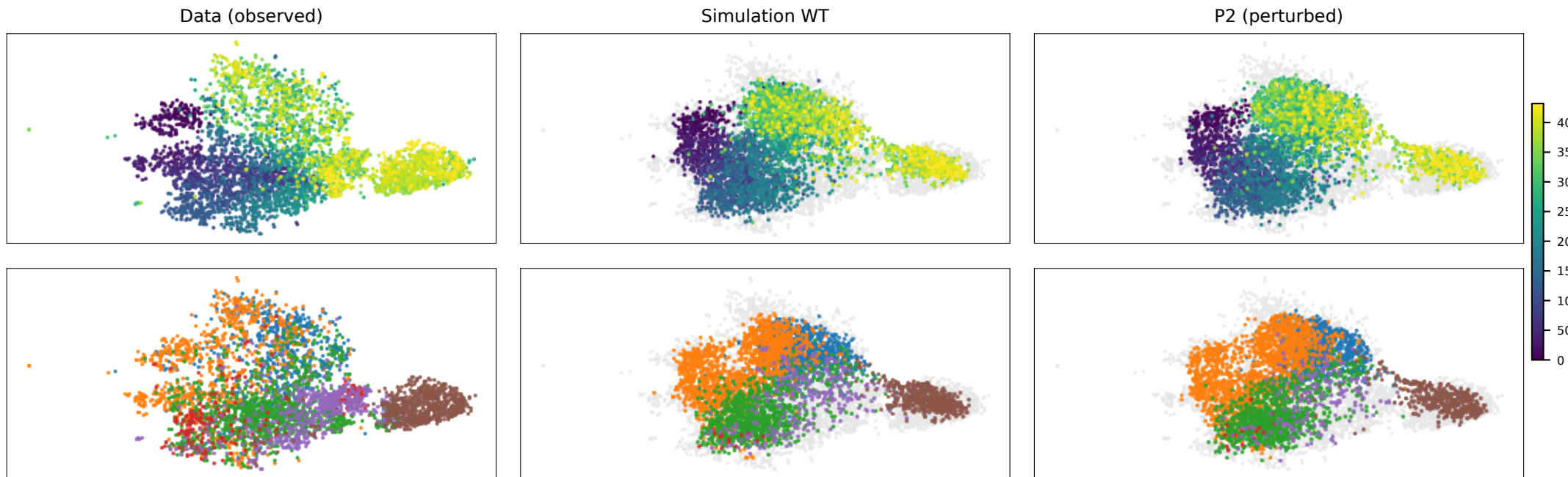
3. Perturbation P1 — OV Obox6

KO_none_OV_Obox6. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



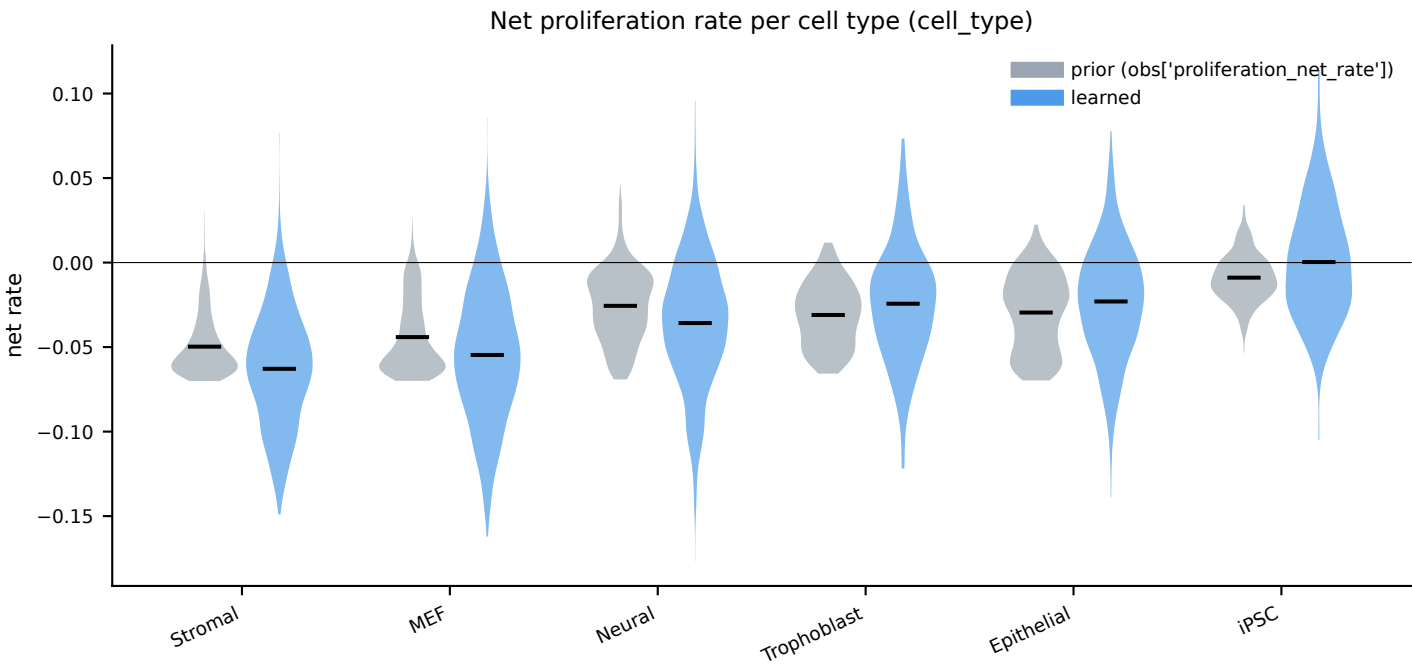
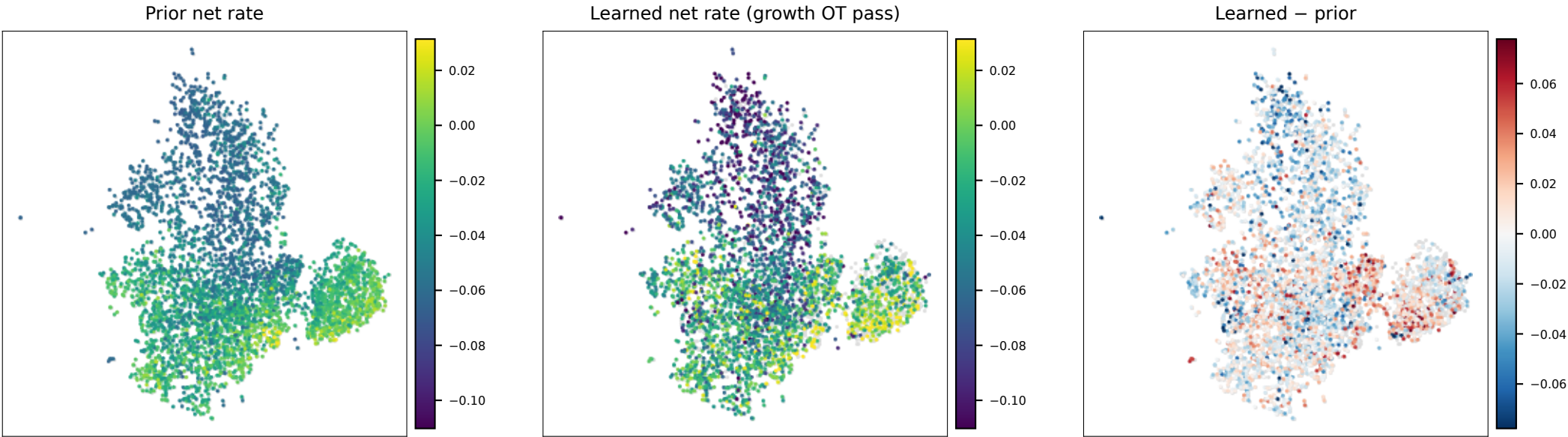
3. Perturbation P2 — OV Zfp42

KO_none_OV_Zfp42. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



4. Proliferation — prior vs learned net rate (birth – death)

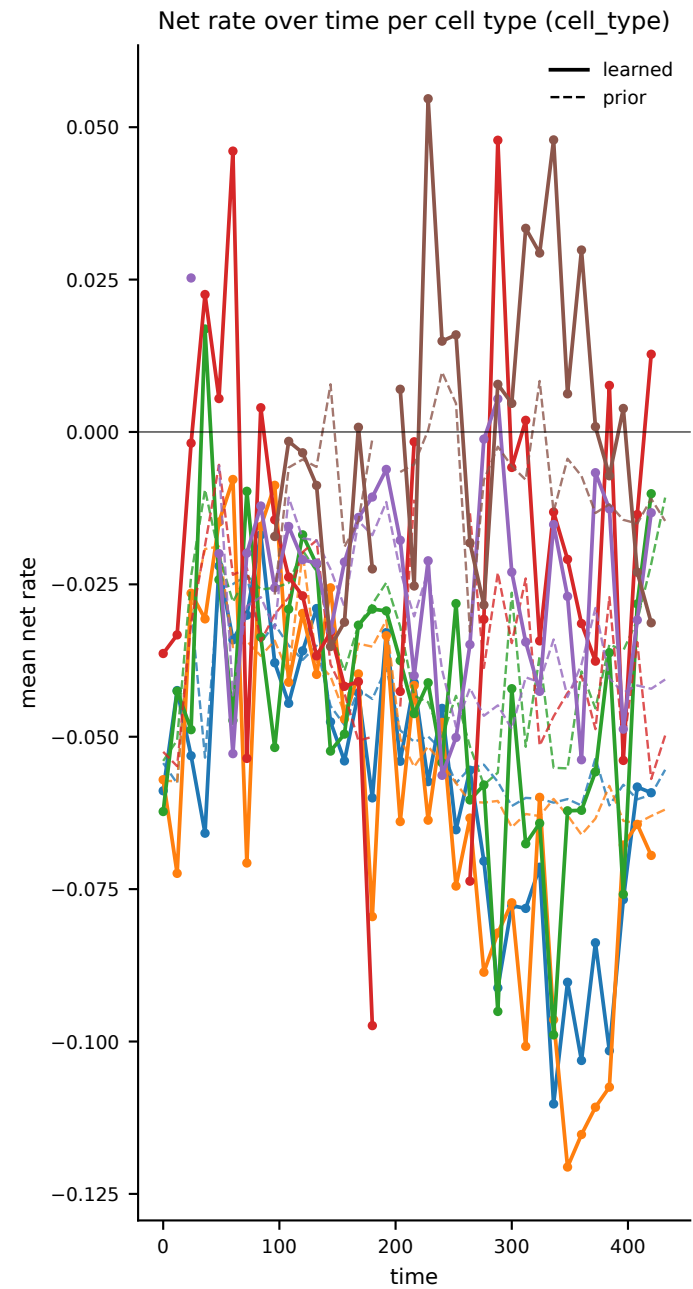
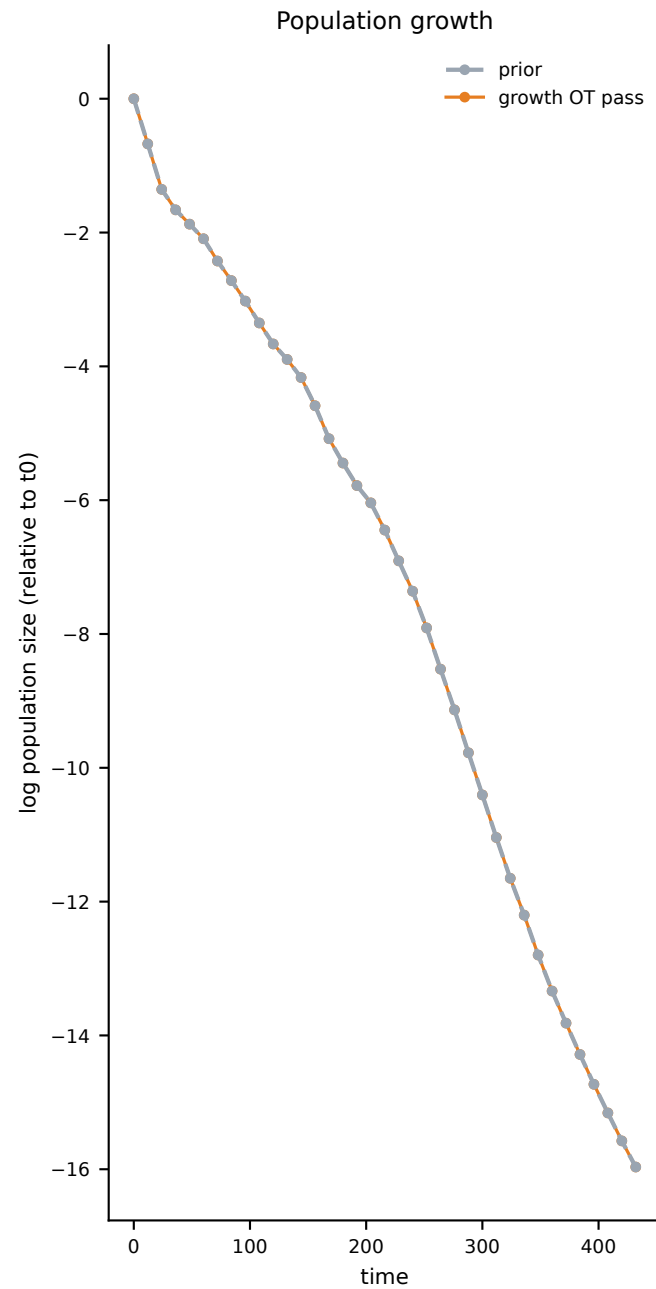
Learned = growth OT pass, mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.



Mean net rate per cell type (cell_type)			
Cell type	Anchor	Prior	Learned
Stromal	—	-0.0497	-0.0629
MEF	—	-0.0441	-0.0547
Neural	—	-0.0256	-0.0358
Trophoblast	—	-0.031	-0.0243
Epithelial	—	-0.0296	-0.023
iPSC	—	-0.00893	0.000316

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; unless Data/population_sizes is given, its absolute level follows the prior.

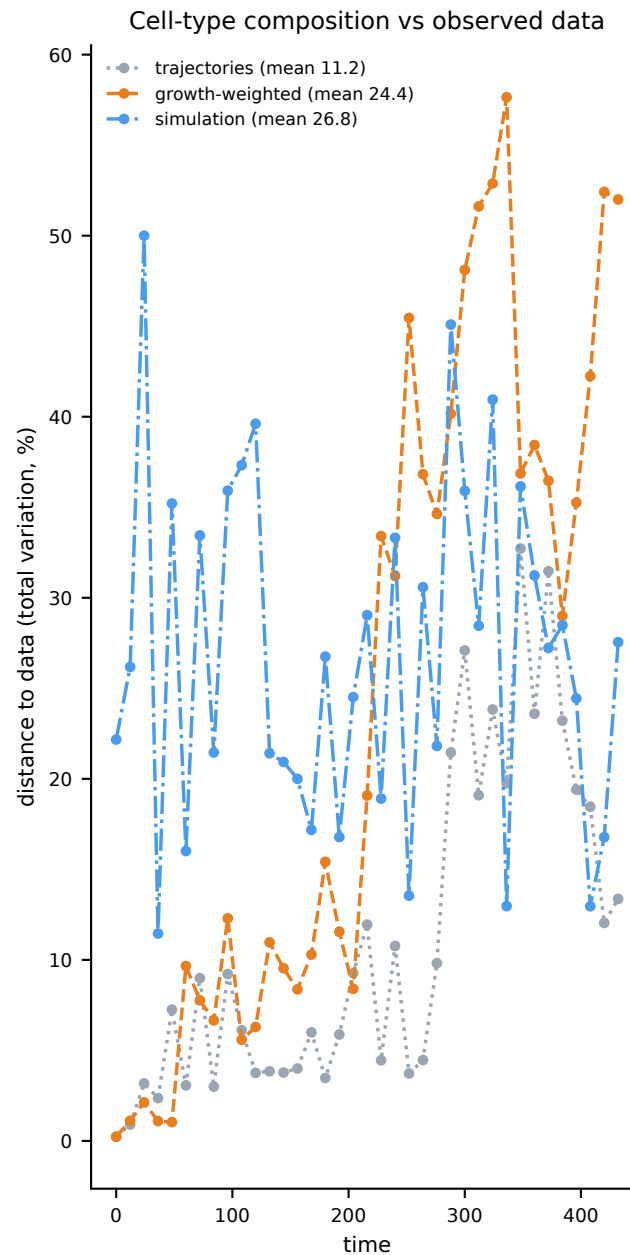
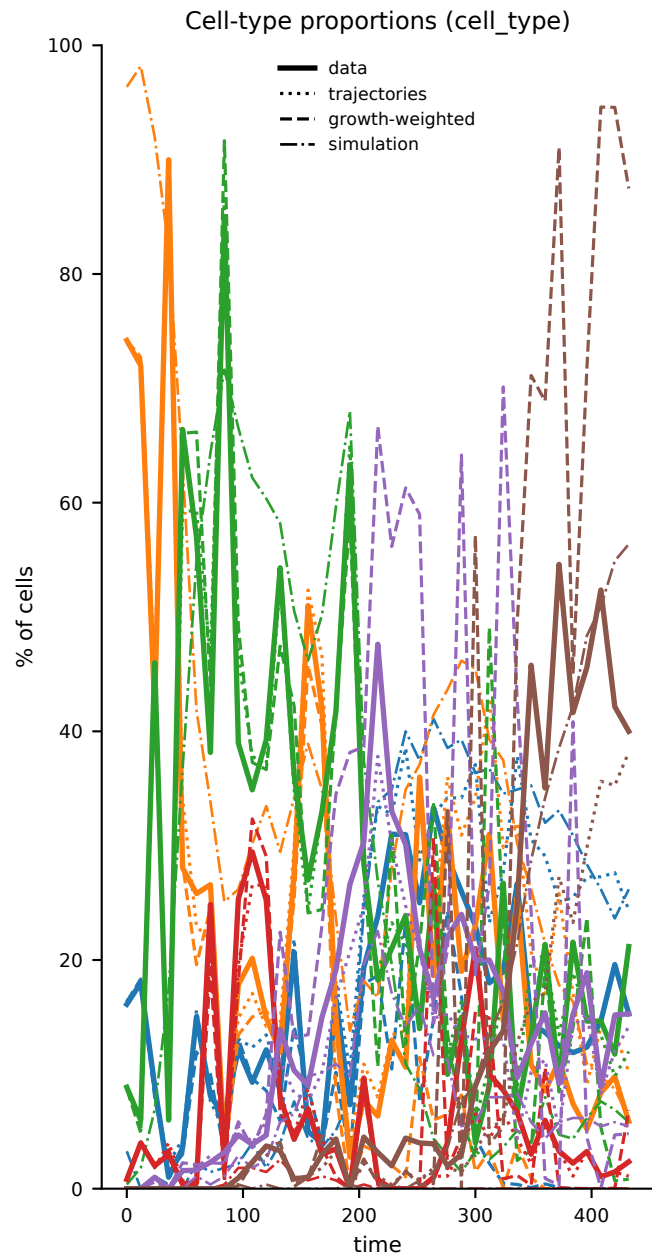


No proliferation MLP
(simulate_with_proliferation = True,
then infer_network_simul)

Stromal MEF Neural Trophoblast Epithelial iPSC

4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion $\exp(JR_{\text{opt}})$ should bring them back to the observed data.

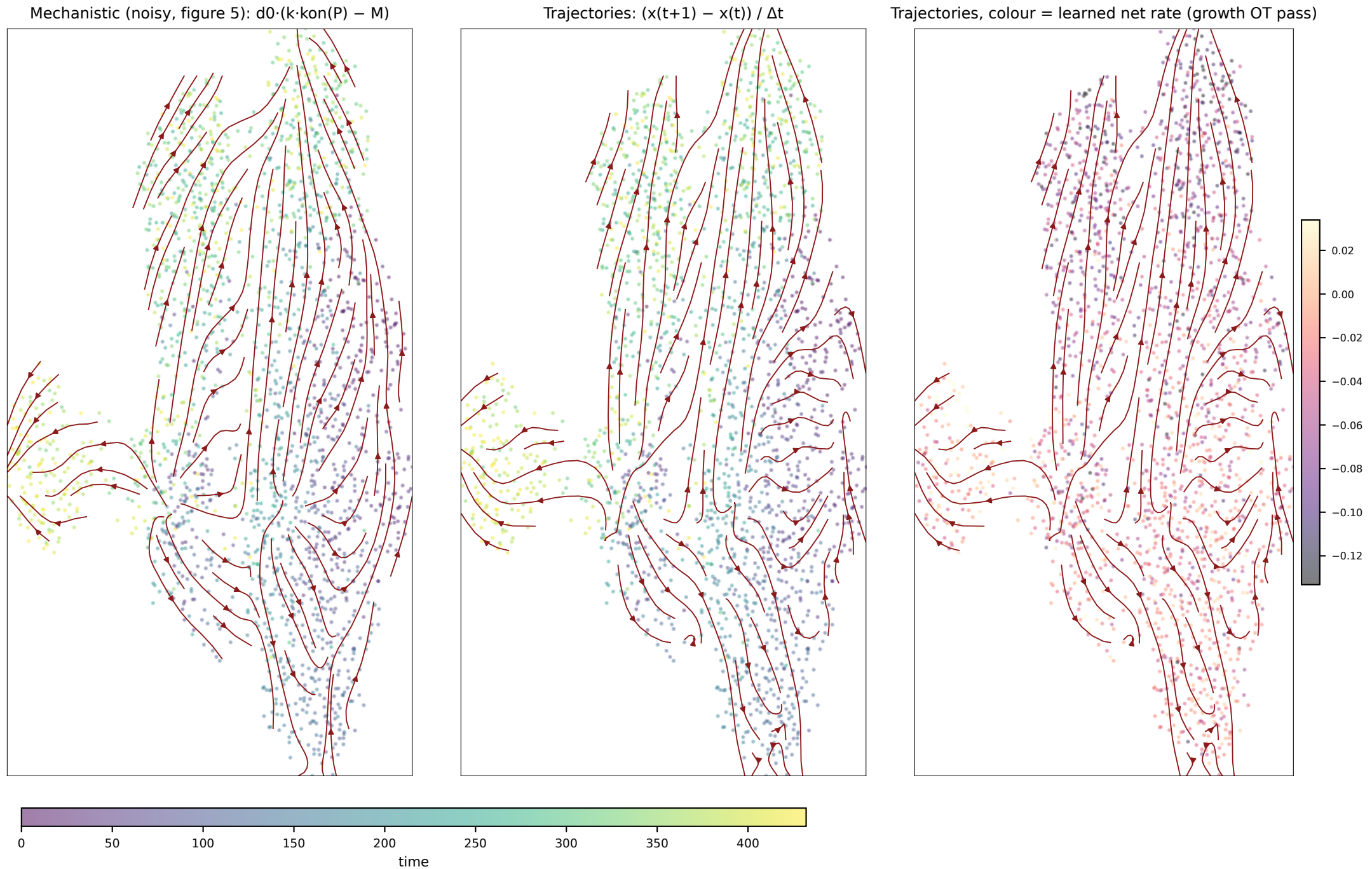


No MLP fit diagnostics
(rerun infer_network_simul
with proliferation)

Stromal MEF Neural Trophoblast Epithelial iPSC

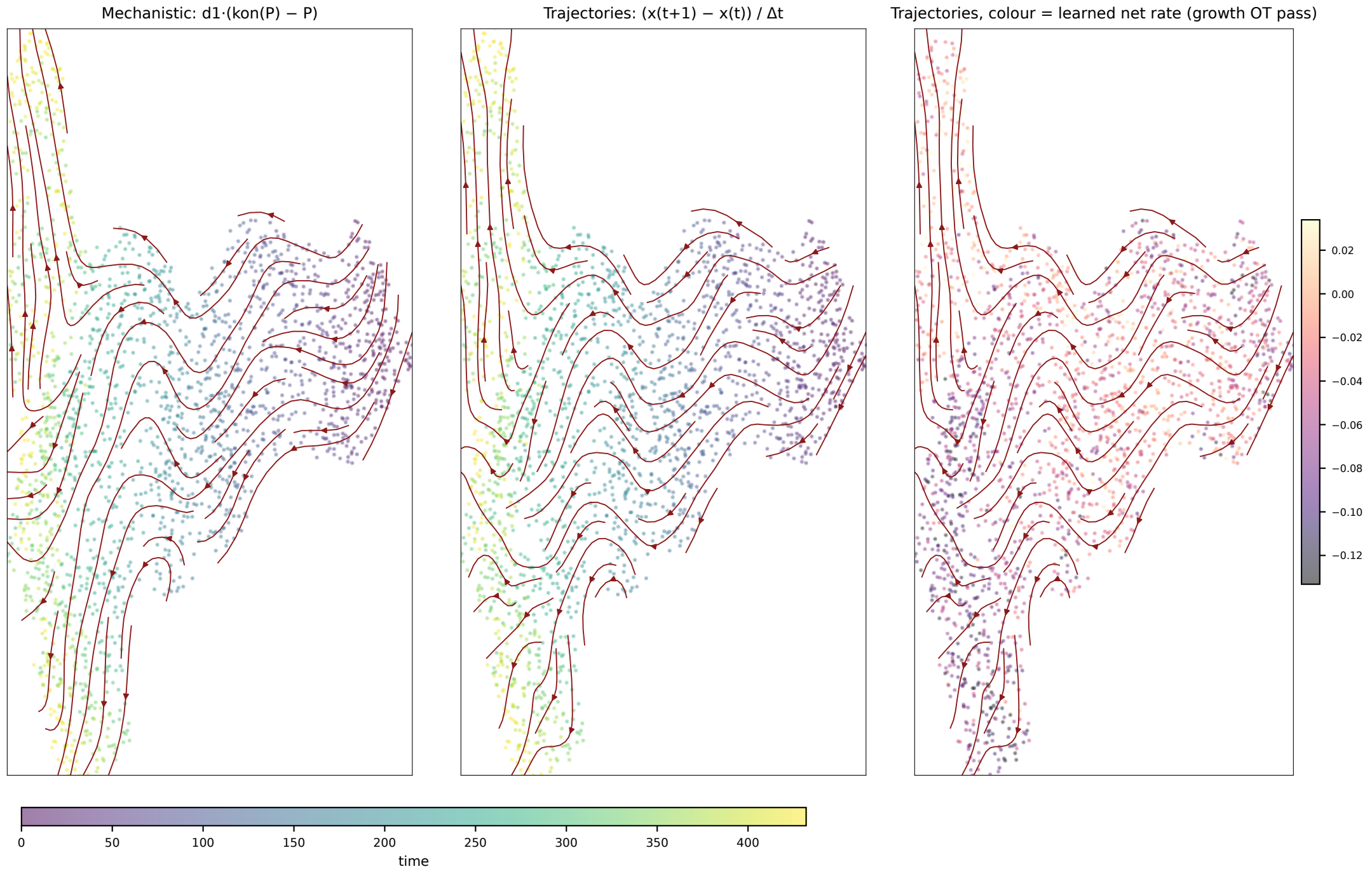
5. Learned dynamics — mRNA velocity fields

1998 trajectory states; kNN-transition projection on a UMAP of log1p NB-sampled mRNA. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.74.



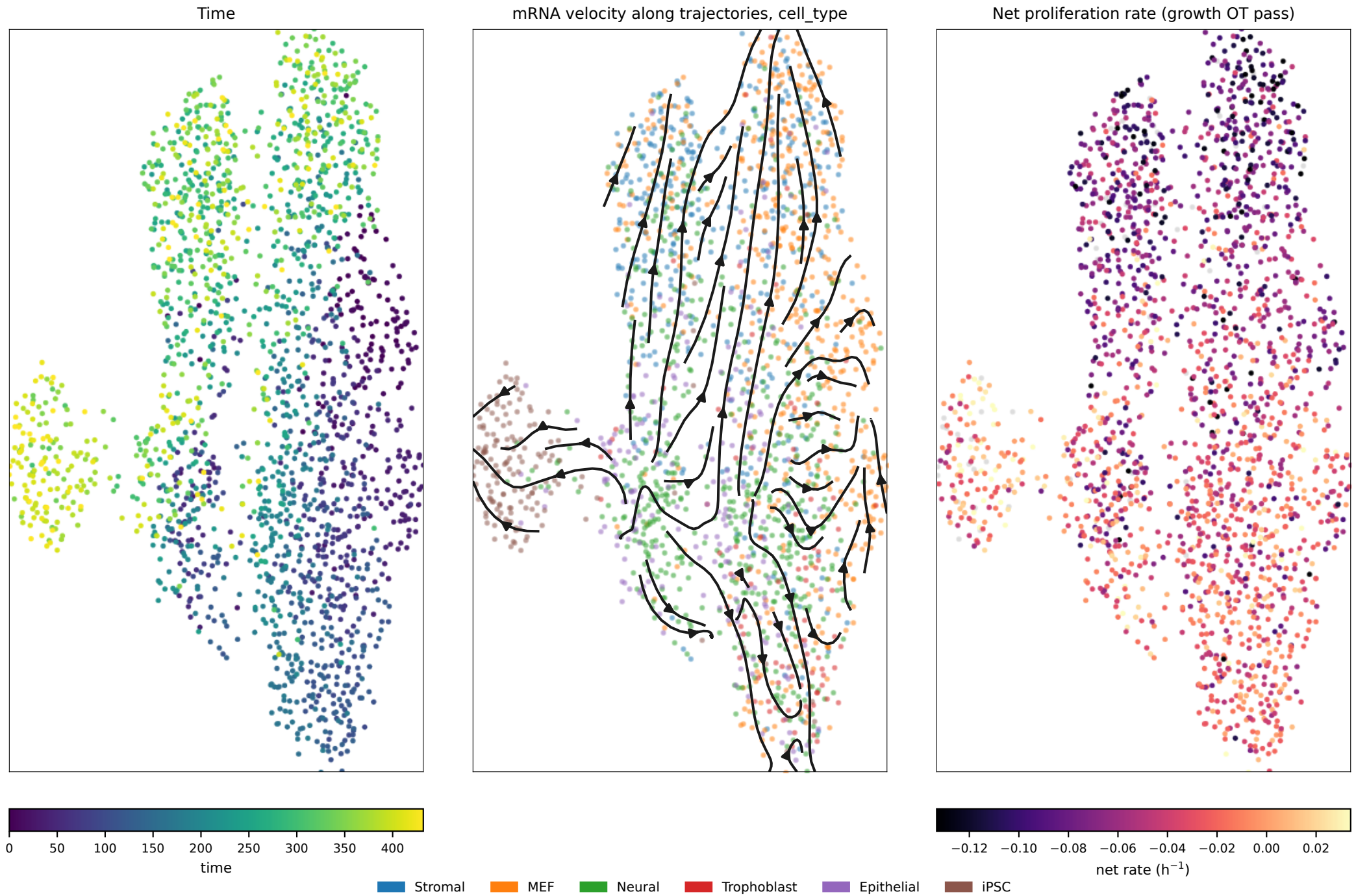
5. Learned dynamics — Proteins velocity fields

1998 trajectory states; kNN-transition projection on a UMAP of protein levels. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.52.



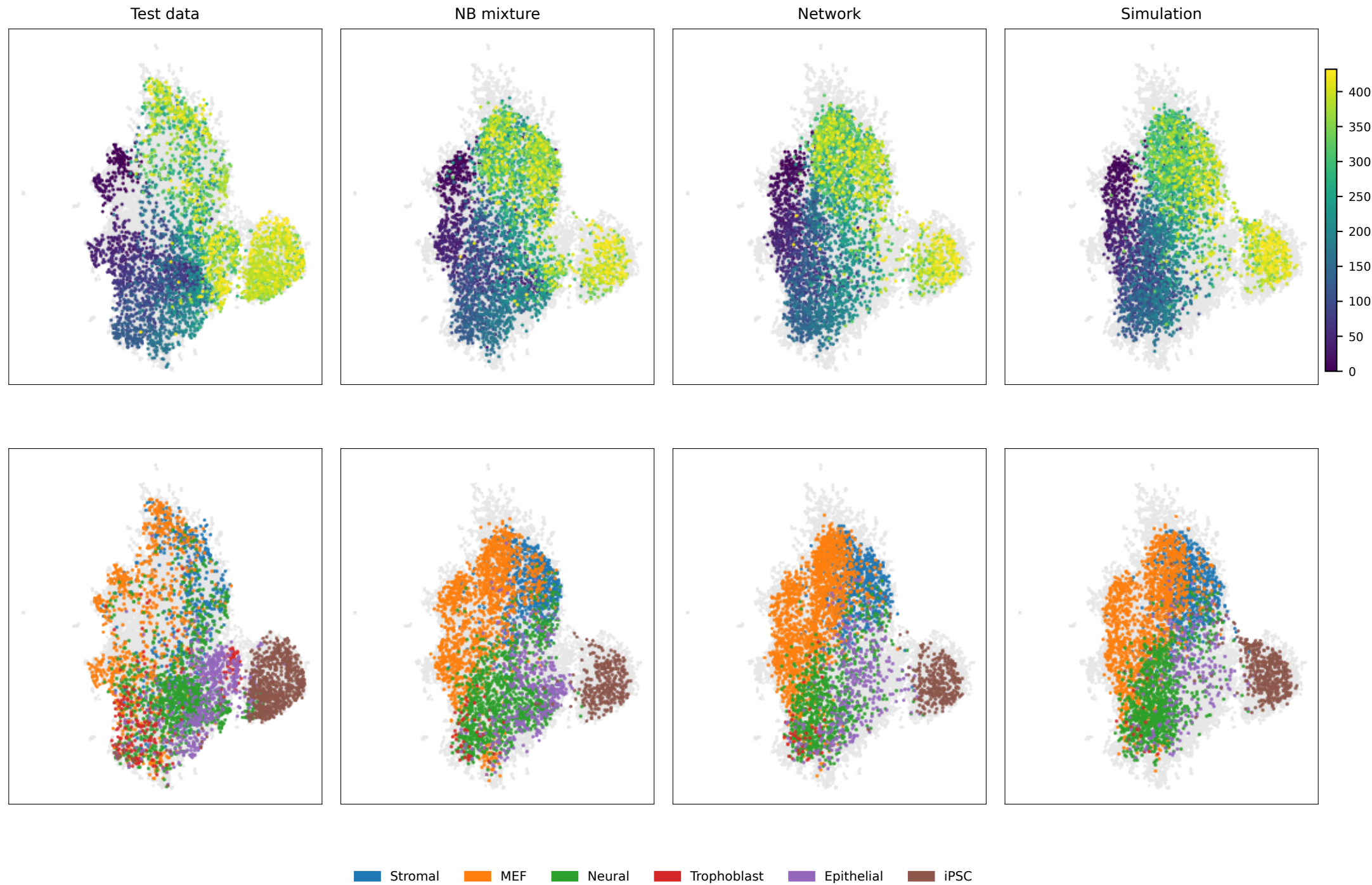
5. Learned dynamics — summary

1998 trajectory states on the UMAP of log1p NB-sampled mRNA; velocity along the inferred trajectories $(x(t+1) - x(t)) / \Delta t$; background of the velocity field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the joint UMAP (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

